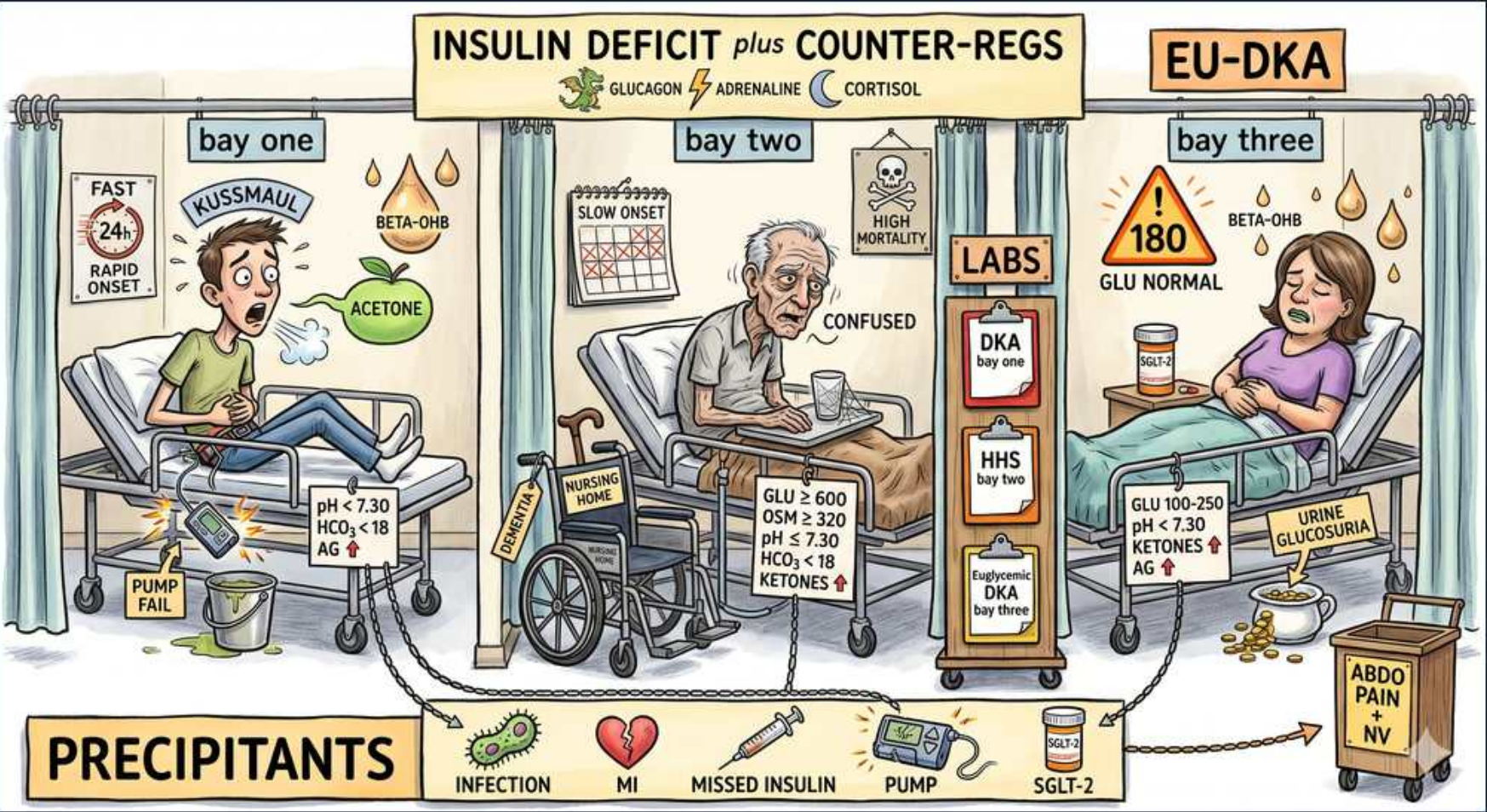


DM — Scene 1 — Acute Hyperglycemic Crisis: DKA vs HHS Diagnosis



1. Insulin deficit + counter-regs banner

WHAT YOU SEE

Banner INSULIN DEFICIT plus COUNTER-REGS (glucagon, adrenaline, cortisol)

WHAT IT ENCODES

Both crises share one engine: insulin deficiency (absolute in DKA, relative in HHS) plus EXCESS counter-regulatory hormones — glucagon, catecholamines (adrenaline), cortisol, and growth hormone. This combination drives hepatic gluconeogenesis/glycogenolysis (→ hyperglycemia) and, when insulin is essentially absent, unrestrained lipolysis and hepatic ketogenesis (→ DKA). In HHS, residual insulin is enough to suppress ketogenesis but not hyperglycemia.

BOARD TRAP

WRONG: 'HHS occurs because the patient has NO insulin at all.' If insulin were truly absent they would ketose (DKA). Discriminator: HHS = relative/partial insulin (enough to block ketones); DKA = near-absolute deficiency (ketogenesis unleashed).

2. DKA fast onset (bay one)

WHAT YOU SEE

Bay one FAST <24h RAPID ONSET young patient

WHAT IT ENCODES

DKA develops rapidly, typically over <24 hours, classically in younger patients with T1DM (but increasingly seen in T2DM 'ketosis-prone' and on SGLT2i). The fast tempo reflects how quickly ketoacidosis accumulates once insulin is withdrawn or a stressor hits. Contrast with HHS, which simmers over days.

BOARD TRAP

WRONG: 'A gradual, days-long decline over a week is the typical DKA course.' That tempo fits HHS. Discriminator: DKA = hours (rapid); HHS = days (insidious).

3. Kussmaul breathing

WHAT YOU SEE

KUSSMAUL deep breathing patient

WHAT IT ENCODES

Kussmaul respirations = deep, rapid, labored breathing that is the respiratory (Winter's) compensation for the metabolic acidosis of DKA — the lungs blow off CO2 to raise pH. Expected PaCO2 ≈ 1.5×HCO3 + 8 (±2). Present in DKA, ABSENT in pure HHS (no significant acidosis).

BOARD TRAP

WRONG: 'Kussmaul breathing in a hyperglycemic patient points to HHS.' Kussmaul means acidosis = DKA. Discriminator: deep compensatory breathing belongs to DKA, not HHS.

4. Acetone / fruity breath

WHAT YOU SEE

ACETONE breath cloud, beta-OHB

WHAT IT ENCODES

Ketogenesis produces three ketone bodies — beta-hydroxybutyrate (dominant in DKA, the one to MEASURE in serum), acetoacetate (what urine dipstick/nitroprusside detects), and acetone (volatile, exhaled → fruity breath). As DKA resolves, beta-OHB converts to acetoacetate, so the urine test can transiently worsen even as the patient improves.

BOARD TRAP

WRONG: 'A negative or only mildly positive urine ketone test rules out significant DKA.' The dipstick misses beta-OHB and underestimates severity. Discriminator: measure serum beta-hydroxybutyrate, not urine acetoacetate, to gauge DKA severity.

5. DKA labs (bay one)

WHAT YOU SEE

DKA card: GLU >250, pH <7.30, HCO3 <18, AG raised, ketones

WHAT IT ENCODES

DKA diagnostic triad: glucose >250 mg/dL, metabolic acidosis (pH <7.30, HCO3 <18 mEq/L), and ketonemia/ketonuria with an ELEVATED anion gap (Na – [Cl + HCO3] > 12). Severity is graded by pH/HCO3/mental status (mild pH 7.25-7.30; severe pH <7.0 with obtundation), NOT by the glucose level.

BOARD TRAP

WRONG: 'The higher the glucose, the more severe the DKA.' Severity is defined by the degree of acidosis and mental status, not glucose. Discriminator: a glucose of 300 with pH 6.9 is far worse than a glucose of 600 with pH 7.28.

6. HHS slow onset (bay two)

WHAT YOU SEE

Bay two SLOW ONSET confused elderly patient, high mortality

WHAT IT ENCODES

HHS evolves insidiously over days to weeks, classically in ELDERLY T2DM patients — often those with limited access to water (nursing home, dementia, post-stroke) who cannot keep up with osmotic-diuresis losses. Mortality (~10-20%) exceeds DKA's (~1-5%), driven largely by age, comorbidities, and the precipitant.

BOARD TRAP

WRONG: 'HHS is less dangerous than DKA because there's no acidosis.' HHS has HIGHER mortality. Discriminator: absence of acidosis does not mean lower risk — HHS kills more often due to severe dehydration, age, and precipitants.

7. HHS altered mental status

WHAT YOU SEE

CONFUSED elderly patient

WHAT IT ENCODES

The hallmark of HHS is profound altered mental status — ranging from confusion to coma — driven by severe hyperosmolarity and dehydration (cellular dehydration of neurons). The depth of AMS correlates with the effective serum osmolality; frank coma usually requires osm >320-340 mOsm/kg. Focal deficits and seizures can occur and may mimic stroke.

BOARD TRAP

WRONG: 'Expect deep coma from ketoacidosis itself in HHS.' HHS has minimal ketosis; the obtundation comes from hyperosmolarity, not acidosis. Discriminator: AMS severity tracks osmolality in HHS — if a DKA patient is comatose, suspect another cause or extreme osmolar/acidotic derangement.

8. HHS labs

WHAT YOU SEE

HHS card: GLU >600, OSM >320, pH >7.30, HCO₃ >18, minimal ketones

WHAT IT ENCODES

HHS diagnostic criteria: glucose >600 mg/dL, effective serum osmolality >320 mOsm/kg [2×Na + glucose/18], pH >7.30, HCO₃ >18, and minimal/absent ketones (no significant ketoacidosis). A mild metabolic acidosis can coexist (lactate from hypoperfusion or some ketosis) — overlap 'mixed DKA-HHS' pictures exist.

BOARD TRAP

WRONG: 'A patient with glucose 700, osm 340, and a large anion gap with ketones has pure HHS.' Significant ketoacidosis means DKA or a mixed picture, not pure HHS. Discriminator: pure HHS has minimal ketones and pH >7.3 — prominent ketoacidosis excludes the 'pure' label.

9. Euglycemic DKA (bay three)

WHAT YOU SEE

Bay three EU-DKA, GLU 180 GLU NORMAL, beta-OHB

WHAT IT ENCODES

Euglycemic DKA = ketoacidosis with near-normal glucose (typically <250, often 150-200). Causes: SGLT2 inhibitors (urinary glucose loss masks hyperglycemia — hold the drug ~3-4 days before surgery/fasting), pregnancy (increased glucose utilization), prolonged starvation, and heavy alcohol use. The acidosis/ketosis/anion gap are the same as classic DKA.

BOARD TRAP

WRONG: 'Glucose is 180 and the patient is on empagliflozin — DKA is excluded.' This is the textbook missed euglycemic DKA. Discriminator: in an acidotic patient on an SGLT2i, the normal glucose is a red herring — check the gap and ketones.

10. Eu-DKA labs

WHAT YOU SEE

GLU 100-250, pH <7.30, ketones, AG raised

WHAT IT ENCODES

Euglycemic DKA labs: glucose 100-250 mg/dL but pH <7.30, positive ketones, and an elevated anion gap — i.e., full ketoacidosis without marked hyperglycemia. Treat exactly as DKA (fluids, K⁺, insulin) but start dextrose-containing fluids EARLY so insulin can keep running to clear ketones without causing hypoglycemia.

BOARD TRAP

WRONG: 'Because glucose is only 200, give fluids alone and withhold insulin.' Insulin is required to stop ketogenesis; just co-administer dextrose. Discriminator: euglycemic DKA still needs insulin — pair it with D5/D10, don't omit it.

11. Urine glucosuria / abdo pain

WHAT YOU SEE

URINE GLUCOSURIA and ABDO PAIN N/V tags

WHAT IT ENCODES

Once glucose exceeds the renal threshold (~180 mg/dL), glucosuria drives an osmotic diuresis (polyuria, dehydration, electrolyte loss). DKA also commonly presents with nausea, vomiting, and diffuse abdominal pain that can mimic an acute surgical abdomen; the pain correlates with acidosis severity and typically resolves as the anion gap closes.

BOARD TRAP

WRONG: 'Diabetic with severe abdominal pain and vomiting — rush to surgery/CT for an acute abdomen.' DKA-related pain usually resolves with correction of acidosis. Discriminator: pain that clears as the gap closes was metabolic; persistent pain after resolution warrants a surgical search.

12. Precipitant: infection

WHAT YOU SEE

Bacteria INFECTION at PRECIPITANTS banner

WHAT IT ENCODES

Infection is the MOST COMMON precipitant of both DKA and HHS (pneumonia and UTI lead). Stress hormones from sepsis raise glucose and counter-regulatory drive. Note: a mildly elevated WBC is expected from stress/dehydration alone, so fever, bacteremia, or a clear source — not just leukocytosis — should prompt cultures and empiric antibiotics.

BOARD TRAP

WRONG: 'The elevated WBC in this DKA patient confirms infection and mandates broad antibiotics for everyone.' Mild leukocytosis is nonspecific in DKA. Discriminator: look for fever/left shift/a source — DKA itself causes a stress leukocytosis without infection.

13. Precipitant: MI

WHAT YOU SEE

Heart MI symbol

WHAT IT ENCODES

Myocardial infarction/ischemia is a key precipitant — and may be SILENT/painless in diabetics due to autonomic neuropathy. Always obtain an ECG and troponin in older patients presenting in hyperglycemic crisis. The MI both triggers the crisis (catecholamine surge) and is worsened by it (demand ischemia from dehydration/tachycardia).

BOARD TRAP

WRONG: 'No chest pain, so an MI can be excluded as the trigger.' Diabetic MIs are frequently asymptomatic. Discriminator: screen with ECG/troponin regardless of chest-pain history in an older patient with DKA/HHS.

14. Precipitant: missed insulin

WHAT YOU SEE

MISSED INSULIN syringe

WHAT IT ENCODES

Insulin omission/nonadherence is a LEADING precipitant of DKA, especially in young T1DM patients (psychosocial stress, eating disorders, cost/access). A common pitfall: patients incorrectly stop their insulin when ill or not eating ('sick-day' error) — basal insulin must NEVER be fully held, even with reduced intake, or DKA ensues.

BOARD TRAP

WRONG: 'A sick T1DM patient who isn't eating should hold all insulin to avoid hypoglycemia.' Stopping basal insulin precipitates DKA. Discriminator: during illness, continue basal insulin and check ketones/glucose frequently — never zero out insulin.

15. Precipitant: pump fail / SGLT2

WHAT YOU SEE

PUMP and SGLT2 symbols

WHAT IT ENCODES

Insulin pump failure (kinked/dislodged catheter, occlusion) causes rapid DKA because pumps deliver only rapid-acting insulin with no long-acting depot — any interruption removes all basal coverage within hours. SGLT2 inhibitors cause EUGLYCEMIC DKA (glucose may be near-normal); hold them ~3-4 days before surgery or during acute illness/fasting.

BOARD TRAP

WRONG: 'A pump-treated patient can't develop DKA quickly because the pump is automated.' A pump occlusion leaves zero basal depot and triggers DKA fast. Discriminator: pump patients in DKA → suspect mechanical failure; on SGLT2i → suspect euglycemic DKA despite normal glucose.
